

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Built and maintained **ML-enabled production services** using **Python**, **scikit-learn**, and **FastAPI** to operationalize data science models; deployed inference services on **AWS Lambda** and **EC2** with automated **CloudWatch** monitoring, reducing model inference latency by **35%** and improving reliability SLAs from **95%** to **99.2%**.
- Developed **ML data pipelines** and **ETL workflows** using **Python (pandas, NumPy)**, **SQL (PostgreSQL, Oracle)**, and **AWS** services; designed **feature engineering** processes for batch and real-time inference, handling **10M+ daily records** with **sub-100ms latency** for production AI systems.
- Supported deployment and operation of **ML models in production environments** through **CI/CD pipelines (Jenkins, Git)**, implementing automated testing, **model versioning**, and **monitoring dashboards** using **Power BI** and **Dynatrace**; reduced model debugging time by **40%** through structured **observability** and **logging practices**.
- Collaborated with data scientists to translate **statistical models** and **machine learning algorithms** into working production solutions; implemented **model evaluation metrics**, **hyperparameter optimization**, and **A/B testing frameworks**, increasing model accuracy by **12%** on held-out validation sets.
- Debugged and improved existing **AI systems** to ensure reliability and performance; analyzed **distributed data flows** across **microservices**, identified bottlenecks in **data pipelines**, and refactored **inference service** code using **async Python** patterns, reducing P1 incident response time by **50%**.
- Applied **engineering best practices** including **code review**, **unit testing**, **documentation**, and **version control** across all AI/ML deliverables; integrated **responsible AI** principles, **data governance**, and **security standards** into model development workflows, ensuring compliance with data protection and production readiness requirements.

SKILLS

AI & Machine Learning – Python, scikit-learn, pandas, NumPy, model deployment, inference services, feature engineering, model evaluation metrics, hyperparameter optimization, A/B testing, statistical analysis, supervised learning, unsupervised learning.

Data Engineering & Pipelines – ETL workflows, data pipelines, batch processing, real-time data flows, feature engineering, data preparation, SQL (PostgreSQL, Oracle, MySQL), distributed systems concepts.

Production AI Systems – ML model deployment, monitoring, observability, debugging, reliability, performance optimization, model versioning, CI/CD (Jenkins, Git).

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, shell scripting, infrastructure monitoring.

Software Engineering – Python, FastAPI, Flask, REST APIs, code review, unit testing, documentation, version control, engineering best practices, security standards.

Data Visualization & Analytics – Power BI, Tableau, DAX, data modelling, business intelligence, dashboard development, Power Query.

Governance & Compliance – Responsible AI, data governance, data protection, security practices, production readiness standards.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Real-Time Fraud Detection ML Service (HCL Technologies)

- Architected end-to-end **ML pipeline** ingesting **50M+ daily transactions** via **AWS Lambda** and **Kinesis**, performing **real-time feature engineering** using **Python** and **pandas** to extract behavioral, temporal, and merchant signals; trained **gradient-boosted** fraud classifier achieving **94.2% precision** and **89.5% recall** on hold-out test set.
- Deployed **inference service** using **FastAPI** and **containerized with Docker**, scaled on **AWS EC2** with auto-scaling groups to handle **10K requests/sec** with **45ms p95 latency**; integrated **model serving** with **Jenkins CI/CD** pipelines and **Git** version control to enable **zero-downtime deployments**.
- Built **monitoring and observability** stack using **CloudWatch**, **Dynatrace**, and **custom Power BI dashboards** tracking model drift, prediction distribution shifts, and inference latency; reduced false-positive fraud alerts by **23%** through real-time **model performance debugging** and retraining workflows.
- Collaborated with data science team to implement **A/B testing framework** for model experimentation; evaluated **5 candidate ML algorithms** (**Random Forest**, **XGBoost**, **LightGBM**) using **scikit-learn** and **cross-validation**, quantifying business impact of **USD 2.3M in prevented fraud losses** annually.
- Established **responsible AI** guardrails including **fairness audits**, **explainability analysis**, **data governance**, and **security compliance** checks; documented **model behavior**, **assumptions**, and **limitations** for regulatory and audit teams; ensured **GDPR-compliant** handling of customer transaction data.

Payments Operations ML Data Platform (HCL Technologies – Banking Domain)

- Engineered **ETL pipelines** processing **SWIFT message flows** (MT/MX formats) and **ISO 20022 message structures** using **Python**, **SQL (PostgreSQL)**, and **AWS**; transformed unstructured payment operations data into curated **feature tables** for **downstream ML models** and operational analytics, ingesting **2M+ payment records daily**.
- Built **data pipeline orchestration** using **Jenkins** scheduled jobs and **cron automation**; implemented **data validation** checks and **schema versioning** to ensure **99.95% data quality** across interconnected **payments systems** and **ERP platforms**, supporting **SEPA**, instant payments, and cross-border payment settlement.
- Created **production monitoring dashboards** in **Power BI** and **Dynatrace** tracking payment message latency, throughput, and transaction reconciliation metrics; enabled **SLA compliance reporting** and operational alerting for **P1/P2 incidents**, reducing mean-time-to-resolution (MTTR) by **30%** for payments operations teams.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)